

CMPE 485 Spring 26 – Term Project

Introduction to Game Programming
Phase I: Topic Selection and Roadmap

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1. Selected Option

For this term project , I have chosen **Option B (Simple Game)**.

2. Game Concept & Mechanics

Title Idea: Cyber-Dash (Working Title)

Genre: First-Person Perspective (FPP) Parkour Action.

Game Mechanics: The core of the game revolves around high-speed movement and quick reflexes. The player will navigate through a 3D environment using mechanics such as wall-running, mid-air dashing, and jumping to complete the level as fast as possible. This aligns with the requirement to formally define the game mechanics.

- **Architecture:** A Finite State Machine (FSM) will be implemented to handle the complex and rapid state changes of the player.
- **Assets:** Visual and audio elements will be sourced for free from the Unity Asset Store, ensuring no pay-to-win elements. Apart from these visual and audio elements, all software will be written from scratch.

3. Proposed Roadmap

This roadmap outlines the development process leading up to the final submission.

- **Step 1 - Core Mechanics & FSM:** Program the fundamental FPP controller physics. Implement the Player FSM to handle running, jumping, and dashing.
- **Step 2 - Advanced Movement & Environment:** Implement wall-detection and wall-running mechanics. Design a test environment using free 3D assets.
- **Step 3 - Technical Challenges:** Iteratively code the game and define three specific technical challenges encountered as the code progresses.
- **Step 4 - Performance Evaluation:** Conduct an in-depth performance evaluation using the Unity profiler. The evaluation will observe the effect of the technically challenging parts on FPS and CPU/GPU usage based on a maximum of three chosen parameters.
- **Step 5 - Final Polish (May 10):** Finalize the report, clean up the code, record the short gameplay video, and prepare for the final submission.